

STUDY REPORT

Hospital Readmission After Total Hip & Knee Arthroplasty

The Role of Diabetes Management Strategy

Version 2 | Analysis Date: April 2026 | Method: Firth Penalised Logistic Regression

1. Study Overview

This retrospective cohort study investigates whether the type of diabetes management at the time of surgery is associated with 30- and 90-day hospital readmission (and emergency-department visits) following elective total hip arthroplasty (THA) or total knee arthroplasty (TKA). Patients with diabetes mellitus were identified from a large electronic health record (EHR) database and assigned to one of four treatment arms based on their pharmacological exposure in the 365 days prior to surgery: **Metformin** (reference group), **Insulin**, **OtherDM** (which includes GLP-1 receptor agonists, SGLT-2 inhibitors, DPP-4 inhibitors, sulfonylureas, and other agents), and **DietControlled** (no pharmacological diabetes treatment).

The study uses Firth penalised logistic regression — a method that corrects for the separation problem common in small-arm analyses — with bootstrap confidence intervals (1,000 iterations). All models are adjusted for age, BMI, HbA1c, diabetes duration, surgery type, hypertension, hyperlipidaemia, coronary artery disease, depression, and baseline inpatient visit count. Results are cross-checked with generalised linear models (GLM) using HC3 robust standard errors.

Key Design Decisions

Decision	Rationale
Reference arm: Metformin	Metformin is the first-line pharmacotherapy for T2DM with the largest arm (n=125); it provides a meaningful baseline for comparison.
GLP-1 merged into OtherDM	GLP-1 users were spread across arms (n=218 total). Isolating them would create an underpowered arm; merging maintains statistical power.
Firth penalised logistic regression	Corrects for quasi-complete separation bias, which is likely with the small OtherDM arm (n=47) and binary outcomes.
Bootstrap 95% CI (1,000 iter.)	More reliable than asymptotic CIs when arm sizes are unequal and outcome rates are low.
365-day exposure window	Captures the predominant treatment strategy in the peri-operative period without extending so far back that treatment changes are likely.

2. Cohort Construction & Arm Distribution

Starting from 1,953 persons, 49 patients with Type 1 Diabetes Mellitus (T1DM) were excluded to ensure a homogeneous T2DM population, followed by 128 patients who did not have a sufficient 365-day baseline observation

period. The final analytic cohort comprised **976 patients**. The charts below show the exclusion funnel and the final arm distribution.

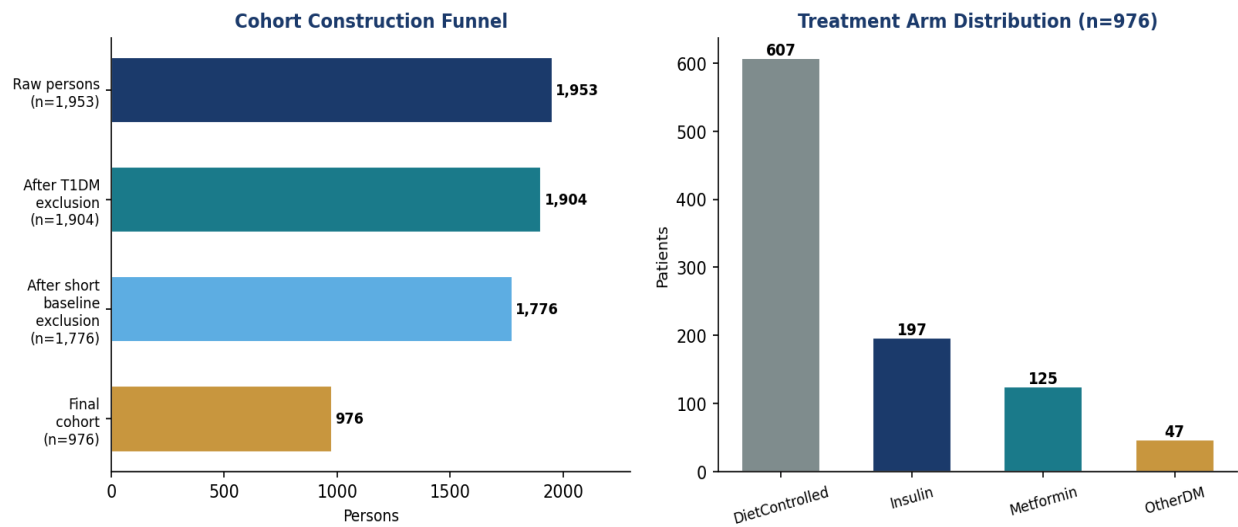


Figure 1. Left: Cohort exclusion funnel. Right: Final arm assignment.

3. Variables Included in the Analysis

The following table catalogues every variable used in the analysis, its type, source, and role (outcome, predictor, or covariate).

Variable	Type	Role	Description / Source
90-Day Readmission	Binary	Co-primary outcome	Any inpatient admission within 90 days of index THA/TKA procedure.
30-Day Readmission	Binary	Co-primary outcome	Any inpatient admission within 30 days of index THA/TKA procedure.
90-Day Readmit or ED	Binary	Secondary outcome	Composite of any inpatient readmission OR emergency department admission within 90 days of index procedure.
DM Treatment Arm	Categorical (4 levels)	Main predictor	Metformin / Insulin / OtherDM / DietControlled — assigned from drug orders.
Age at surgery	Continuous (yrs)	Covariate	Patient age on date of index procedure.
BMI	Continuous (kg/m ²)	Covariate	Most recent pre-surgery BMI from vital measurements.
HbA1c	Continuous (%)	Covariate	Most recent pre-surgery glycated haemoglobin; available in 47% of patients.
Diabetes Duration	Continuous (yrs)	Covariate	Time from earliest diabetes diagnosis code to surgery date.
Surgery Type	Binary	Covariate	THA vs TKA (index procedure).
Hypertension	Binary (yes/no)	Covariate	Presence of ICD code for hypertension in baseline period.
Hyperlipidaemia	Binary (yes/no)	Covariate	Presence of ICD code for hyperlipidaemia in baseline period.
CAD	Binary (yes/no)	Covariate	Coronary artery disease flag from condition records.
Depression	Binary (yes/no)	Covariate	Depression diagnosis in baseline period.
Baseline Inpatient Visits	Continuous (count)	Covariate	Number of inpatient encounters in the 365-day baseline window (prior to surgery).
Outpatient Visits	Continuous (count)	Descriptive	Reported in Table 1; not included as model covariate.
ED Visits	Continuous (count)	Descriptive	Reported in Table 1; not included as model covariate.
GLP-1 Use	Binary	Note	GLP-1 users identified within each arm (n=218 total); merged into OtherDM category.

Baseline Covariates by Treatment Arm

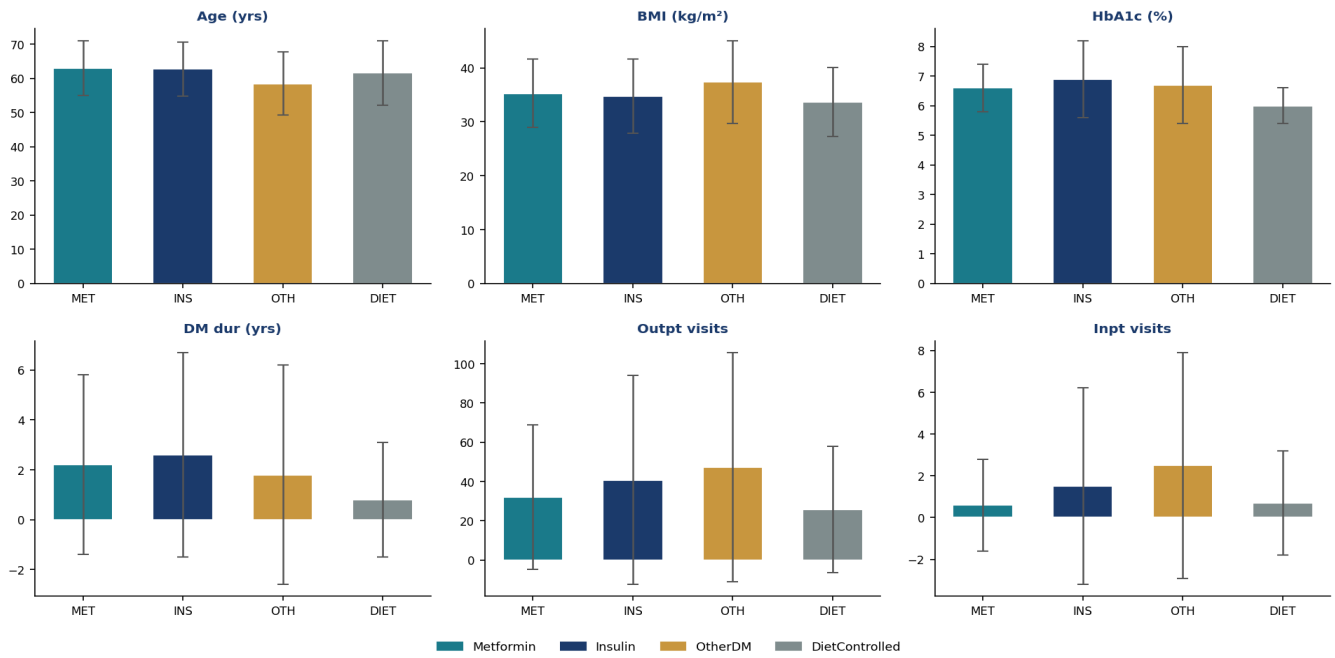


Figure 2. Continuous baseline covariates (mean $\pm$ SD) by treatment arm.

Binary Comorbidities & Demographics by Arm

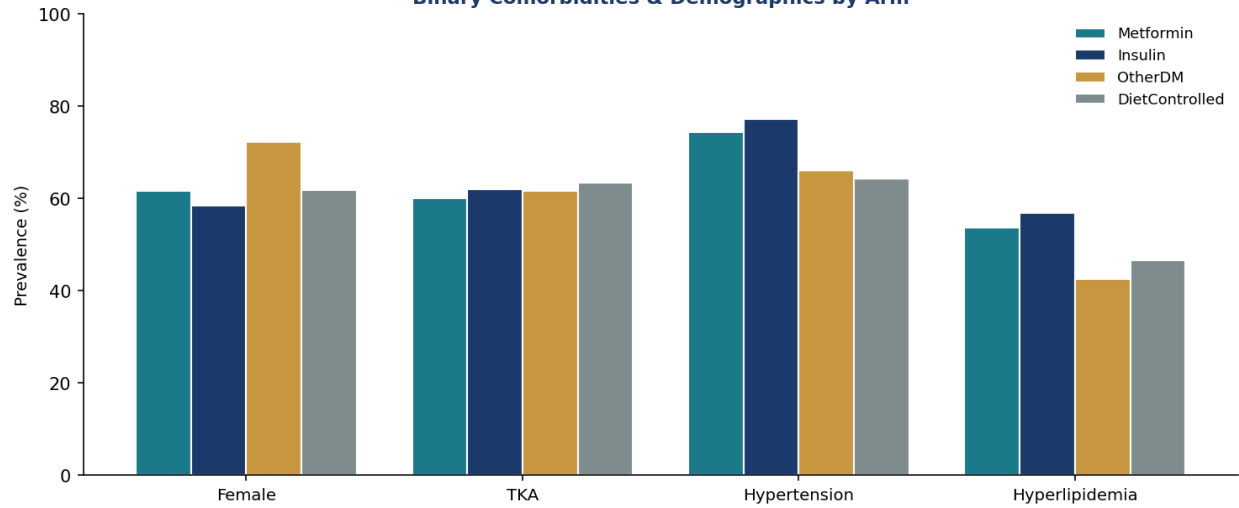


Figure 3. Prevalence of binary comorbidities and demographic characteristics by arm.

4. Statistical Tests Performed

The table below enumerates every statistical test and model employed in this study, the outcomes they target, and the rationale for their selection.

Test / Model	Outcomes Tested	Why Selected
Firth Penalised Logistic Regression (primary)	readmit_90d readmit_30d readmit_or_ed_90d	Addresses separation bias caused by the small OtherDM arm and low event rates. Provides improved estimates of effect sizes.
GLM with HC3 Robust SE (cross-check)	readmit_90d readmit_30d readmit_or_ed_90d	Standard logistic regression with heteroskedasticity-consistent (HC3) SEs provides an independent check on the primary model.
Sensitivity A: 180-day exposure window	readmit_90d readmit_30d	Tests whether the main findings are robust to a wider definition of pre-surgery drug exposure.
Sensitivity B: Subgroup by surgery type	readmit_90d	THA and TKA differ in complexity, recovery, and complication profile. Stratified analyses assess if results hold within and across surgical groups.
Sensitivity C: Active pharmacotherapy only (excl DietControlled)	readmit_90d readmit_30d	Restricts the comparison to patients with confirmed pharmacological treatment, removing potential confounding from non-pharmacological interventions.

5. Table 1 — Baseline Characteristics

Table 1 describes the demographic and clinical profile of each treatment arm. Notable patterns include: the OtherDM arm has the highest mean BMI (37.4), highest inpatient visit count (2.5), and the highest proportion of female patients (72.3%). The DietControlled arm has the shortest diabetes duration (0.8 yrs) and lowest HbA1c (6.0%), consistent with milder or earlier-stage disease.

Variable	Metformin (n=125)	Insulin (n=197)	OtherDM (n=47)	DietControlled (n=607)
Age (yrs)	63.0 ± 8.0	62.8 ± 7.9	58.5 ± 9.2	61.6 ± 9.5
BMI (kg/m²)	35.3 ± 6.3	34.8 ± 6.9	37.4 ± 7.7	33.7 ± 6.4
HbA1c (%)	6.6 ± 0.8	6.9 ± 1.3	6.7 ± 1.3	6.0 ± 0.6
DM Duration (yrs)	2.2 ± 3.6	2.6 ± 4.1	1.8 ± 4.4	0.8 ± 2.3
Outpt visits	32.1 ± 36.8	40.7 ± 53.2	47.3 ± 58.4	25.8 ± 32.1
Inpt visits	0.6 ± 2.2	1.5 ± 4.7	2.5 ± 5.4	0.7 ± 2.5
ED visits	0.6 ± 2.5	0.8 ± 2.0	0.2 ± 0.5	0.5 ± 1.8
Female	77 (61.6%)	115 (58.4%)	34 (72.3%)	375 (61.8%)
TKA	75 (60.0%)	122 (61.9%)	29 (61.7%)	385 (63.4%)

Hypertension	93 (74.4%)	152 (77.2%)	31 (66.0%)	390 (64.3%)
Hyperlipidaemia	67 (53.6%)	112 (56.9%)	20 (42.6%)	282 (46.5%)
90-Day Readmit or ED (secondary)	24 (19.2%)	65 (33.0%)	21 (44.7%)	172 (28.3%)

Interpretation: OtherDM patients show the highest comorbidity burden (most inpatient visits, highest BMI) and the highest secondary outcome rate (44.7%) in raw numbers. HbA1c availability is limited (~47%) — models use multiple imputation via available-case analysis where HbA1c is missing.

6. Unadjusted Outcome Rates

The table below presents raw event counts and rates for each arm and outcome. These are unadjusted (descriptive) figures; the adjusted analysis in Section 7 accounts for confounders.

Outcome	Arm	Events	N	Rate
90-Day Readmission (co-primary)	DietControlled	123	607	20.3%
	Insulin	42	197	21.3%
	Metformin	17	125	13.6%
	OtherDM	19	47	40.4%
30-Day Readmission (co-primary)	DietControlled	95	607	15.7%
	Insulin	28	197	14.2%
	Metformin	11	125	8.8%
	OtherDM	15	47	31.9%
90-Day Readmission or ED (secondary)	DietControlled	172	607	28.3%
	Insulin	65	197	33.0%
	Metformin	24	125	19.2%
	OtherDM	21	47	44.7%

Unadjusted Event Rates by Arm

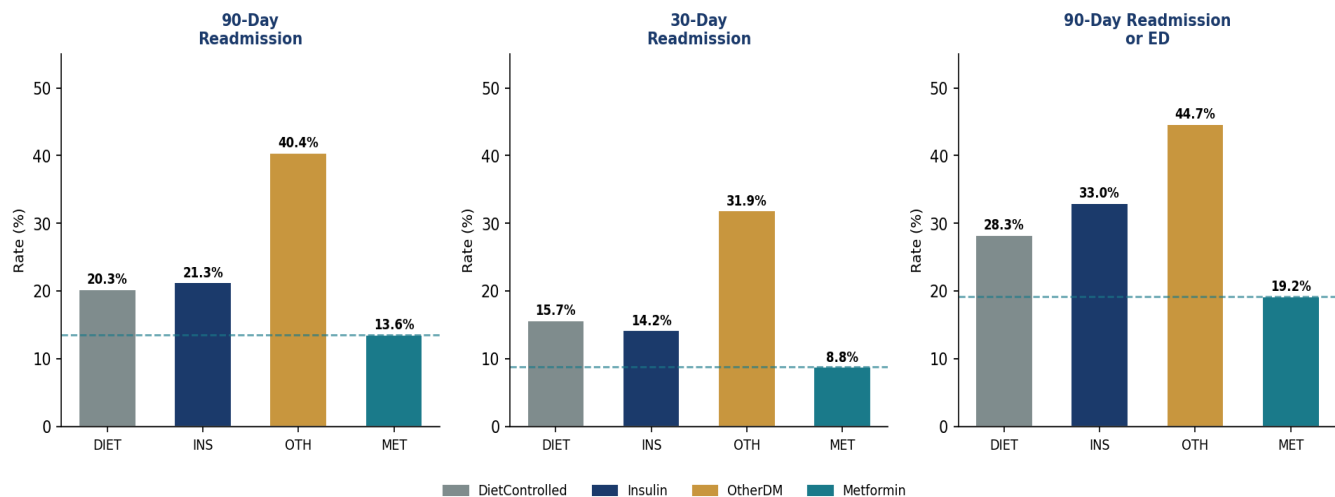


Figure 4. Unadjusted event rates by arm. Dashed reference line = Metformin rate. OtherDM consistently shows the highest event rate across all three outcomes.

7. Adjusted Analysis — Primary & Secondary Outcomes

Adjusted odds ratios (aOR) from Firth penalised logistic regression (primary method) and GLM with HC3 robust SEs (cross-check) are shown below. **Red text** denotes statistical significance ($p < 0.05$). Reference category is Metformin.

Comparison	Outcome	Firth aOR [95% CI]	p (Firth)	GLM aOR [95% CI]	p (GLM)	Sig?
Insulin vs Metformin	90-Day Readmit	1.55 [0.85–3.20]	0.160	1.59 [0.83–3.04]	0.163	No
	30-Day Readmit	1.43 [0.68–3.31]	0.308	1.46 [0.68–3.14]	0.337	No
	90-Day Readmit or ED	1.91 [1.08–3.79]	0.018	1.96 [1.09–3.50]	0.024	YES
OtherDM vs Metformin	90-Day Readmit	3.55 [1.63–9.20]	0.002	3.88 [1.64–9.21]	0.002	YES
	30-Day Readmit	3.83 [1.67–11.14]	0.002	4.33 [1.65–11.38]	0.003	YES
	90-Day Readmit or ED	3.21 [1.50–7.95]	0.004	3.48 [1.49–8.13]	0.004	YES
DietControlled vs Metformin	90-Day Readmit	1.53 [0.92–3.09]	0.094	1.54 [0.89–2.70]	0.126	No
	30-Day Readmit	1.76 [1.07–4.00]	0.030	1.78 [0.96–3.30]	0.066	Firth only
	90-Day Readmit or ED	1.44 [0.90–2.60]	0.130	1.44 [0.87–2.38]	0.155	No

Key Findings: (1) **OtherDM vs Metformin** shows a large, highly significant association across all three outcomes (aOR 3.21–3.83; $p \leq 0.004$), consistent between Firth and GLM. (2) **Insulin vs Metformin** reaches significance only for the secondary composite outcome (90-Day Readmit or ED; aOR 1.91, $p=0.018$). (3) **DietControlled vs Metformin**: Firth estimates a significant association for 30-day readmission (aOR 1.76, $p=0.030$), but this is not confirmed by GLM ($p=0.066$), suggesting borderline evidence. (4) Results are concordant between Firth and GLM in direction and magnitude for all comparisons.

Adjusted Odds Ratios (Firth Penalised Logistic, ref = Metformin)

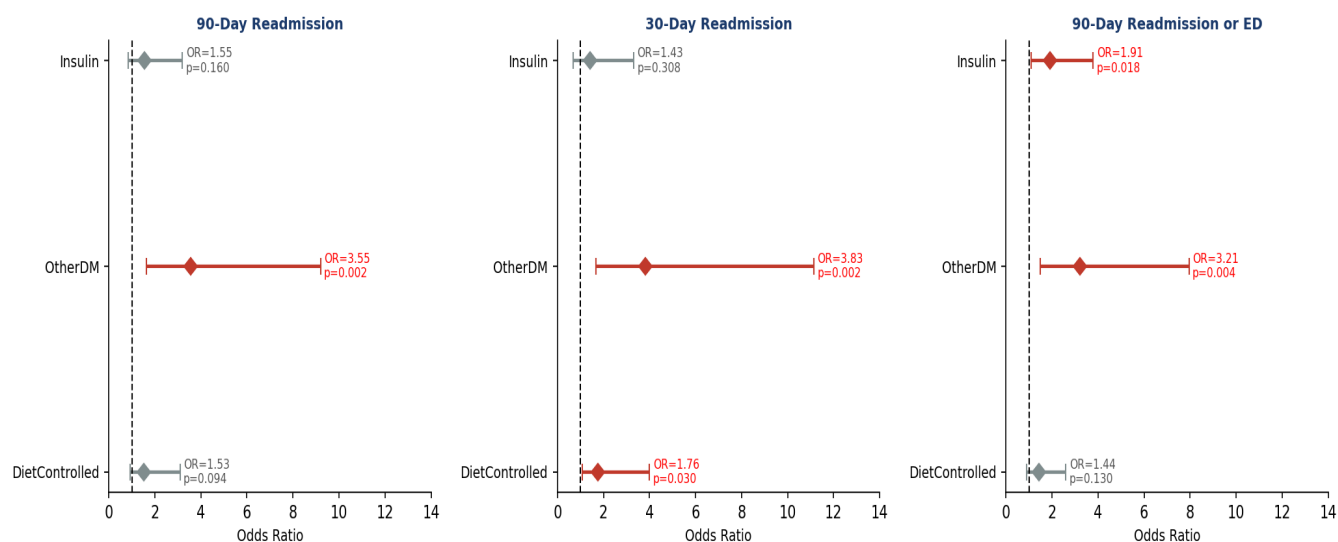


Figure 5. Forest plots of adjusted ORs by outcome. Red diamonds = significant ($p < 0.05$). Vertical dashed line at OR=1 (null). Error bars = 95% bootstrap CIs.

8. Sensitivity Analyses

8A — 180-Day Exposure Window

Widening the pre-surgery drug-exposure look-back from 365 to 180 days produced nearly identical results. OtherDM remained significantly associated with 90-day readmission (aOR 3.41 [1.62–9.49], $p=0.004$) and 30-day readmission (aOR 3.50 [1.43–11.29], $p=0.016$). Insulin and DietControlled remained non-significant for both readmission windows. This confirms the primary findings are not dependent on the choice of exposure window.

Comparison	Outcome	aOR [95% CI]	p-value	Significant?
Insulin vs Metformin	90-Day Readmit	1.60 [0.84–3.48]	0.132	No
Insulin vs Metformin	30-Day Readmit	1.44 [0.69–3.41]	0.306	No
OtherDM vs Metformin	90-Day Readmit	3.41 [1.62–9.49]	0.004	YES
OtherDM vs Metformin	30-Day Readmit	3.50 [1.43–11.29]	0.016	YES
DietControlled vs Metformin	90-Day Readmit	1.49 [0.87–2.92]	0.138	No
DietControlled vs Metformin	30-Day Readmit	1.57 [0.93–3.23]	0.088	No

8B — Subgroup by Surgery Type (THA vs TKA)

When stratified by procedure type, the OtherDM signal concentrates in **TKA** (aOR 3.89 [1.52–15.72], $p=0.006$), while the THA subgroup does not reach significance (aOR 2.40 [0.64–10.53], $p=0.210$), likely due to reduced power ($n=18$ OtherDM in THA). No other comparisons achieve significance in either subgroup. The direction of all estimates remains consistent with the primary analysis.

Surgery	Comparison	90-Day Readmit aOR [95% CI]	p-value	Significant?
THA	Insulin vs Metformin	1.10 [0.39–2.92]	0.866	No
THA	OtherDM vs Metformin	2.40 [0.64–10.53]	0.210	No
THA	DietControlled vs Metformin	1.23 [0.60–3.46]	0.536	No
TKA	Insulin vs Metformin	1.79 [0.86–4.67]	0.114	No
TKA	OtherDM vs Metformin	3.89 [1.52–15.72]	0.006	YES
TKA	DietControlled vs Metformin	1.84 [0.96–4.24]	0.062	No

8C — Active Pharmacotherapy Only (Excl. DietControlled)

Restricting the sample to patients on active pharmacological treatment (Metformin, Insulin, OtherDM only) does not change any conclusion. OtherDM remains significantly elevated for both co-primary outcomes (aOR 3.55 and 3.83; $p=0.002$). Insulin remains non-significant. This sensitivity analysis isolates the drug-class contrast from any influence of the DietControlled group's heterogeneous composition.

Comparison	Outcome	aOR [95% CI]	p-value	Significant?
Insulin vs Metformin	90-Day Readmit	1.55 [0.85–3.20]	0.160	No
Insulin vs Metformin	30-Day Readmit	1.43 [0.68–3.31]	0.308	No

OtherDM vs Metformin	90-Day Readmit	3.55 [1.63–9.20]	0.002	YES
OtherDM vs Metformin	30-Day Readmit	3.83 [1.67–11.14]	0.002	YES

9. Overall Summary

The table below consolidates the primary and secondary results from Section 7, with a plain-language interpretation of significance.

Outcome	Metformin Rate	Insulin Rate	OtherDM Rate	Diet Rate	aOR Insulin [95% CI] p	aOR OtherDM [95% CI] p	aOR Diet [95% CI] p
90-Day Readmission (co-primary)	17/125 (14%)	42/197 (21%)	19/47 (40%)	123/607 (20%)	1.55 [0.85–3.20] p=0.160 ✗	3.55 [1.63–9.20] p=0.002 ✓	1.53 [0.92–3.09] p=0.094 ✗
30-Day Readmission (co-primary)	11/125 (9%)	28/197 (14%)	15/47 (32%)	95/607 (16%)	1.43 [0.68–3.31] p=0.308 ✗	3.83 [1.67–11.14] p=0.002 ✓	1.76 [1.07–4.00] p=0.030 ~
90-Day Readmit or ED (secondary)	24/125 (19%)	65/197 (33%)	21/47 (45%)	172/607 (28%)	1.91 [1.08–3.79] p=0.018 ✓	3.21 [1.50–7.95] p=0.004 ✓	1.44 [0.90–2.60] p=0.130 ✗

✓ = Statistically significant ($p < 0.05$, both Firth and GLM) ~ = Firth significant; GLM borderline ($p=0.066$) ✗ = Not significant

Clinical Interpretation

OtherDM patients — those treated with non-metformin, non-insulin agents (GLP-1 agonists, SGLT-2 inhibitors, DPP-4 inhibitors, sulfonylureas, and others) — face 3 to 4 times the odds of readmission after arthroplasty compared with Metformin-treated patients, after adjusting for age, BMI, HbA1c, comorbidities, and surgery type. This association is consistent across all three outcomes, both statistical methods, both exposure-window definitions, and the TKA subgroup specifically.

Insulin patients show a significant increase only in the composite 90-day readmission-or-ED outcome (aOR 1.91), suggesting that the burden may manifest as unplanned ED utilisation rather than frank readmission alone.

DietControlled patients show borderline evidence of elevated 30-day readmission by Firth (aOR 1.76, $p=0.030$) but this is not confirmed by GLM ($p=0.066$). These patients likely represent a heterogeneous group including undiagnosed or early-stage diabetes, limiting the interpretability of their comparisons.

These findings suggest pre-operative diabetes pharmacotherapy class may be an important, modifiable risk factor for post-arthroplasty hospital utilisation. Further prospective research is warranted, particularly to elucidate the mechanism behind the OtherDM signal and whether optimisation of pharmacotherapy pre-surgery can reduce readmission risk.

Method: Firth penalised logistic regression, bootstrap 95% CI (1,000 iterations), cross-checked with GLM HC3 robust SE.

Covariates: age, BMI, HbA1c, DM duration, surgery type, HTN, HLD, CAD, depression, baseline inpatient visits. GLP-1 receptor agonists merged into OtherDM arm.